Process Characterization Plan

A-Mab Drug Substance — Low-pH Viral Inactivation (Step 6) — PCP-006

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Field	Value
Document ID	PCP-006
Document class	Process Characterization Plan
Title	Process Characterization Plan: Low-pH Viral Inactivation (Step 6)
Product	A-Mab
Version	2.0
Effective date	2026-09-30
Supersedes	PCP-006 Version 1.0 (2026-01-05)
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor	Novacyte Biologics
Development site	Cambridge, MA
Commercial drug substance site	Grafton, WI

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Approvals

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2026-01-05	Initial issue
2.0	2026-09-30	Reissued under INV-001 and INV-002: a priori intermediate acceptance criteria, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
ALCOA+	attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available
AMD	analytical method decision
AMV	analytical method validation
BLA	Biologics License Application
CE-SDS	capillary electrophoresis with sodium dodecyl sulfate
CEX	cation exchange chromatography, an analytical method for charge variants
CPP	critical process parameter
CQA	critical quality attribute
DS	drug substance
GPP	general process parameter
HMW	high molecular weight species
IAC	intermediate acceptance criterion
ICH	International Council for Harmonisation
icIEF	imaged capillary isoelectric focusing
KPP	key process parameter
LMW	low molecular weight species
LRF	log reduction factor
MSAT	Manufacturing Science and Technology
MVM	minute virus of mice
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
PAR	proven acceptable range
PRESS	prediction error sum of squares
QA	Quality Assurance
SEC	size exclusion chromatography
SOP	standard operating procedure
TCID50	50 % tissue culture infectious dose

Abbreviation	Meaning
WC-CPP	well-controlled critical process parameter
XMuLV	xenotropic murine leukemia virus

1 Purpose and scope

This plan sets out the Stage 1 characterization of the low-pH viral inactivation step, Step 6 of the A-Mab drug substance process (U.S. Food and Drug Administration 2011; CMC Biotech Working Group 2009). It applies the Process Characterization Master Plan PCMP-001 to this step and adds what is specific to the step, which is its scale-down and spiking models, its designs, its sampling plan and the values of its criteria. The studies are run in a scale-down model at the development site in Cambridge, MA. Their results support the operating ranges and the control strategy that the Process Transfer Plan PTP-001 moves to the commercial drug substance facility in Grafton, WI. The commercial process runs at the 15,000 L scale.

Low-pH viral inactivation holds the Protein A eluate at acid pH for a set time and then neutralizes it. The hold inactivates enveloped viruses. The step contributes a log reduction of xenotropic murine leukemia virus (XMuLV) to the modular viral clearance claim of the process (International Council for Harmonisation 2023a). The same conditions act on the antibody and can raise the levels of aggregate, fragments and acidic variants in the pool.

The plan covers the four parameters of the step, which are inactivation pH, hold time, temperature and A-Mab concentration. It covers the attributes that the step sets or can change, the scale-down model and its qualification, and the spiking model for XMuLV. It sets out the designed experiments and the univariate study, the acceptance criteria and the planned analysis, including the proven acceptable ranges (PARs). The Protein A step that produces the load of this step is characterized under PCP-005. The results of this plan are reported in PCR-006, which rolls up into the Process Characterization Master Report PCMR-001.

1.1 Changes in this revision

Version 1.0 of this plan was approved on 2026-01-05, before the first study of the step started on 2026-01-07. Version 2.0 is a protocol amendment issued under the QA investigation INV-001. QA opened INV-001 on 2026-08-03 after its review of PCMR-001 Version 1.0 for BLA readiness, and closed it on 2026-09-18 with the classification major. The investigation found that the criteria used to judge each step in the PAR analysis were in no approved protocol and had been set after the data were seen (INV-001 finding 1). Corrective action CA2 of INV-001 amends each step plan with the criteria and the fragment method. This version is that amendment for Step 6, and it makes the following changes.

1. It sets out the intermediate acceptance criteria for the outputs of this step as PCMP-001 Version 2.0 derives them (Section 7.2). It also sets out the rule by which the verification of the platform worst cases may tighten or widen the criteria upstream of this step (Section 7.4). The derivation uses no result of the studies in this plan.
2. It adds fragments as an output of the step and judges them on results by non-reduced capillary electrophoresis with sodium dodecyl sulfate (nrCE-SDS), following the method decision AMD-001 of 2026-03-20 (Section 5.3). Size exclusion chromatography (SEC) reads fragments high in the Protein A eluate and in the neutralized pool (INV-001 finding 6).
3. It qualifies the scale-down model by its offset from manufacturing scale, and it states every judgement of an output with a known offset at manufacturing scale (Section 5.1).
4. It states the tests a response-surface model must pass before a range is set from it (Section 7.1). Version 1.0 asked for a significant model, no significant lack of fit and residuals with no structure. This version adds a limit on how far predicted R^2 may fall short of R^2, a regression of the absolute residuals on the predicted values and of the residuals on run order, and a cut-off on the Cook's distance of a single run.
5. It judges XMuLV clearance at the least favourable combination of inactivation pH, hold time and temperature within their normal operating ranges (NORs), because viral clearance is validated at worst-case conditions (Section 8). Version 1.0 judged XMuLV clearance on a single batch, as it judged the quality attributes.
6. It records the execution window of every study that this plan governs (Table 11.1). The Version 1.0 records gave no execution dates, so the equipment tables of the package appeared to cite expired calibrations (INV-001 finding 5).
7. It records the date on which Version 1.0 was approved. The title block of Version 1.0 carried 2026-07-24, the date on which the Version 1.0 reports were approved. The revision history now gives the approval date of this plan (INV-001 finding 5).

Table 1.1 compares the criteria that the Version 1.0 report applied to the outputs of this step with the criteria of this version.

Table 1.1: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0, low-pH viral inactivation.

Attribute	Version 1.0[a)]		Version 2.0	
	Criterion	Rule	Criterion	Basis
Acidic variants (%)	none[b)]	–	18.0–40.0	Drug substance
Aggregates, HMW (%)	≤ 2.17	R1, margin 6.0	≤ 7.02	Carried back, tightened[c)] from ≤ 10.0
Fragments, LMW (%)	not assessed[d)]	–	≤ 2.60	Carried back
XMuLV clearance (log10)	≥ 4.93	R3	≥ 6.60	Platform claim

[a)] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[b)] Version 1.0 set no step criterion and judged the step against the drug-substance criterion.
[c)] Tightened on verification: a downstream step did worse within its NOR than the platform worst case.
[d)] Version 1.0 did not report fragments. The attribute was added at Version 2.0.
[R1] Version 1.0 rule: nominal back-calculation / margin.
[R3] Version 1.0 rule: requirement − nominal of other steps.

The Version 1.0 aggregate limit was the drug substance criterion carried back through the nominal clearance of this campaign and divided by a margin of 6.0 (INV-001 finding 2). The Version 2.0 aggregate criterion is wider than that limit, and stricter than its own a priori value of 10.0 % HMW (Section 7.2). The Version 1.0 limit was never an approved criterion, because it was applied in the report at analysis and appeared in no approved plan. For XMuLV clearance, Version 1.0 subtracted the nominal clearance of the other steps in this campaign from the cumulative requirement. Version 2.0 holds the step to its platform claim, which is the stricter criterion.

The factors, ranges, NORs, set-points and designs are those of Version 1.0, against which the studies were planned and around which they were run. PCR-006 Version 2.0 moved the set-points of inactivation pH and hold time, moved the NOR of inactivation pH and narrowed the NORs of hold time and temperature. This plan does not adopt those values. The notes to Table 6.1 give them and name PCR-006 as the document that made each change.

2 Objectives

The studies under this plan have the following objectives.

1. To quantify the effects of inactivation pH, hold time and temperature, and of their interactions, on XMuLV clearance, aggregate, fragments and acidic variants.
2. To fit response-surface models that predict these outputs over the range studied, and to test each model against the adequacy criteria of PCMP-001 §7.2.
3. To determine, for each attribute that has a criterion at this step, the PAR of inactivation pH, hold time and temperature, and to establish whether the NOR of each lies within it.
4. To verify the platform worst cases of this step on which the criteria upstream of it rely.
5. To assess the effect of A-Mab concentration on aggregate, acidic variants and XMuLV clearance one factor at a time, and to support its range against the criteria for aggregate and acidic variants.
6. To provide the evidence for the classification of each parameter under SOP-4001, and for the contribution of the step to the control strategy in PCMR-001.

3 Related documents

This plan is one document of the A-Mab characterization package. Table 3.1 lists its parent documents, its paired report, the master report into which it rolls up and the investigation INV-001, under which this version is issued.

Table 3.1: Documents related to this plan.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-006	Process Characterization Report (Low-pH Viral Inactivation (Step 6))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision

Table 3.2 lists the procedures and method validations that this plan cites. The SOP and AMV numbers are placeholders.

Table 3.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-1002	Design, Execution and Statistical Analysis of DoE Studies
	SOP-2009	Operation of the Low-pH Viral Inactivation Step
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
	SOP-3011	Size-Variant Analysis by SEC-HPLC
	SOP-3013	Charge-Variant Analysis by icIEF
	SOP-3015	Purity by Non-Reduced CE-SDS
Analytical method validation	AMV-3017	XMuLV Infectivity Titre (TCID50)
	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3013	Charge Variants (icIEF)
	AMV-3020	Purity (non-reduced CE-SDS)
	AMV-3019	Protein Concentration by A280 (UV)

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Low-pH viral inactivation is the step after Protein A capture and before cation exchange chromatography. The Protein A eluate leaves the column at low pH. It is adjusted to the inactivation pH and held for the hold time at the set temperature. Base is then added to neutralize the pool. The neutralized pool is the load of cation exchange. The step is operated under its standard operating procedure, SOP-2009.

Low pH inactivates an enveloped virus by denaturing the glycoproteins of its envelope and by disrupting the envelope itself. The virus needs the envelope glycoproteins to bind to a cell and to fuse with its membrane. Inactivation follows first-order kinetics. Its rate constant rises as the pH falls and as the temperature rises, and the log reduction is proportional to that rate integrated over the hold time. The inactivation pH is expected to govern the step, because the envelope glycoproteins denature much faster with each small fall in pH near the operating range. A longer hold gives a larger log reduction at any pH and temperature, and a warmer hold speeds the denaturation of the envelope glycoproteins.

The same conditions act on the antibody. At acid pH the second constant domain of the heavy chain of the antibody is partially unfolded, which exposes hydrophobic surface. Molecules with exposed hydrophobic surface associate, so aggregate forms for as long as the pool is held at low pH. Aggregate forms faster in a warmer pool, because the unfolding and the association both speed up with temperature, and in a more concentrated pool (Section 4.3). On the platform, time at low pH adds a small amount of acidic variants. This plan measures that rise in every run of the designs, and PCR-006 verifies it against the platform worst case (Section 7.4). The low-pH hold also causes a small amount of acid-catalysed cleavage in the hinge region, which adds fragments. Over a hold of the length studied, the rise in acidic variants is expected to be small against the level formed in the production bioreactor, and the rise in fragments small against the level the load already carries. Aggregate formed in the hold is carried into cation exchange, the step that removes it.

The low-pH step belongs to the Novacyte downstream platform and was used for three earlier antibodies, X-Mab, Y-Mab and Z-Mab. Their history establishes the mechanism of the step and the range of performance to expect from it (CMC Biotech Working Group 2009). The same history supplies the platform claim for XMuLV clearance and the largest increases in aggregate, fragments and acidic variants that the step has shown. The intermediate acceptance criteria in Section 7.2 are built on these values.

The step is not expected to change the glycan attributes, which are formed in the production bioreactor and which no downstream step changes (PCMP-001 §3). It makes no claim for the parvovirus minute virus of mice (MVM), because a parvovirus has no envelope and so no envelope glycoproteins for low pH to denature. The criteria also assume no change in host cell protein across the step, and no clearance of host cell protein is claimed for it.

4.2 Quality attributes in scope

Table 4.1 lists the attributes in scope of this plan, with their drug substance criteria and their criticality from the register of critical quality attributes (CQAs) in PCMP-001 §3. The register lists every attribute of the drug substance that has a criterion, whatever its criticality. Criticality was assigned with Tool #1 of the A-Mab case study, as the product of impact and uncertainty (CMC Biotech Working Group 2009). For every attribute except viral clearance, a higher Tool #1 score gives an equal or higher level of criticality. XMuLV clearance was assigned very high criticality by rule, because a viral contamination would harm the patient directly, whatever its score (PCMP-001 §3). Attributes of high or very high criticality are called critical.

Table 4.1: Quality attributes in scope of this plan, from the drug substance CQA register.

Attribute	DS criterion	Criticality	Tool #1	Formed or set at
Viral clearance — XMuLV (enveloped)	≥ 16.7 log10 (cumulative)[a)]	VH[b)]	20	Low-pH Viral Inactivation[c)]
Aggregates (HMW)	≤ 5 % HMW (SEC)	H	60	Production Bioreactor
Fragments (LMW, clipped species)	≤ 2 % (nrCE-SDS)	L	12	Harvest and Clarification
Acidic charge variants (deamidation)	18–40 % (CEX/icIEF)	VL	4	Production Bioreactor

[a)] Cumulative requirement, met by the claims of low-pH viral inactivation, anion exchange and virus filtration together.
[b)] Very high by rule for viral clearance, whatever the Tool #1 score.
[c)] The register places the cumulative requirement at this step.
DS, drug substance. Tool #1, the score of impact multiplied by uncertainty (A-Mab case study). VH, very high. H, high. L, low. VL, very low.

The step sets one attribute, XMuLV clearance. The requirement for XMuLV clearance is cumulative, and the step contributes its platform claim to it (Section 7.3). Aggregate and acidic variants are formed in the production bioreactor, and the register assigns fragments to harvest. The low-pH hold can add to each of these three attributes, so each has an intermediate acceptance criterion at the output of this step.

The register gives the criterion for acidic variants against two methods, cation exchange chromatography (CEX) and imaged capillary isoelectric focusing (icIEF).

Acidic variants are measured by icIEF throughout this campaign, under the charge-variant procedure SOP-3013.

4.3 Risk-based prioritization of parameters

Every parameter of this step was assessed in RA-001 before characterization began. Each parameter was ranked on its potential impact on the CQAs and on its potential to interact with other parameters, as ICH Q9 describes (International Council for Harmonisation 2023b). RA-001 gives the scores and the rationale for each assignment.

The inactivation pH, hold time and temperature of the hold were assigned to multivariate study. A lower pH, a longer hold and a higher temperature each raise the log reduction of XMuLV, and each also raises the aggregate that forms in the hold. The three are also expected to interact. The log reduction is the product of a rate that pH and temperature set and the time over which that rate acts. A design that varies the three parameters together is therefore needed to estimate the clearance and the aggregate at their combinations.

The A-Mab concentration was assigned to univariate study. A higher antibody concentration raises the rate at which partially unfolded molecules meet, so aggregate forms faster in a more concentrated pool. The mechanism of the step gives no reason to expect an effect of the antibody concentration on the inactivation of the virus, which acts on the viral envelope. The univariate study measures the log reduction at each setting of the concentration, so it tests that expectation (Section 6.4). The concentration is also a property of the Protein A pool and not a setting of the hold, so it is assessed one factor at a time.

5 Materials and methods

5.1 Scale-down model and its qualification

The scale-down model holds the Protein A eluate in a temperature-controlled vessel at laboratory scale. The eluate is adjusted to the inactivation pH, held and neutralized in that vessel as in manufacture. The parameters of the step do not depend on scale. The inactivation pH, hold time, temperature and A-Mab concentration are therefore set to the same values as in manufacture, and the additions of acid and base are scaled with the volume of the pool.

The qualification runs are made under SOP-1001 in study S-VI-SDM, before the first characterization run. The study compares three runs of the scale-down model at set-point with five manufacturing-scale batches, for aggregate and acidic variants in the neutralized pool. These runs give the offset of each of the two outputs. The decision whether the scale-down model is qualified for an output rests on whether the output meets its criterion at every combination of NOR low, set-point and NOR high, and that judgement needs the response-surface model. The decision is therefore made at the analysis, in PCR-006.

The model is qualified for each output separately, by its offset from manufacturing scale and not by a test of equivalence (PCMP-001 §7.1). The rule has four parts.

1. The offset of an output is the mean of the scale-down runs at set-point minus the mean of the manufacturing-scale batches. It is reported with its two-sided 90 % confidence interval.
2. Every scale-down result for aggregate or acidic variants is translated to manufacturing scale by subtracting the offset before it is compared with a criterion. Ranges and conclusions for these two outputs are therefore stated at manufacturing scale.
3. The model is qualified for an output when the conclusion against its criterion over the NOR is the same with the offset at either end of its confidence interval. The conclusion is whether the output meets its criterion at every combination of NOR low, set-point and NOR high of the three factors of the response-surface design.
4. A scale-down model whose conclusion for an output changes between the two ends of the confidence interval of the offset is not used for that output.

Where the cause of an offset is known, PCR-006 will state it.

Fragments and XMuLV clearance are not compared at both scales, so their offsets are unknown, not zero. These two outputs are judged as the model predicts them, with no translation to manufacturing scale. For fragments, representativeness rests

on the mechanism of the step. Cleavage in the hinge region at low pH depends on pH, time and temperature, which the scale-down model sets to the manufacturing values, and not on the size of the vessel. The plan therefore assumes that the scale-down model forms fragments at the rate of manufacture (Section 12). For XMuLV clearance, representativeness rests on the spiking model.

Virus cannot be introduced into the manufacturing process, so XMuLV clearance is measured in a spiking model of the step (International Council for Harmonisation 2023a). The spiking model is the scale-down model, run on the same Protein A eluate and at the same inactivation pH, hold time and temperature. The XMuLV spike is added to the pool after the pool has been acidified and before the hold begins, at the volume ratio fixed in the operating procedure for the step, SOP-2009. That ratio is small enough that the spike changes neither the pH nor the antibody concentration of the pool. The input titre is taken from a sample of the spiked pool at the start of the hold and the output titre at the end of the hold, and the log reduction is the difference between the two. Each run of a design is made twice. The spiked run gives the two titres, and an unspiked run at the same settings gives the quality attributes.

The inactivation rate depends on the pH and the temperature of the pool and on its matrix, which is the buffer of the Protein A eluate. It does not depend on the volume of the pool. The spiking runs use Protein A eluate produced at the set-points of the Protein A step, so the matrix is that of manufacture, and the runs of the designs hold the A-Mab concentration at its set-point. The small-scale result therefore represents the manufacturing step at that concentration, provided that the whole pool reaches the inactivation pH. The univariate study measures the log reduction across the range of the concentration (Section 6.4).

5.2 Operation

Each run starts from Protein A eluate produced at the set-points of the Protein A step. The eluate is adjusted to the inactivation pH of the run and brought to the temperature of the run. The hold time is counted from the moment the inactivation pH is reached. At the end of the hold the pool is neutralized with base and filtered. Neutralization stops the inactivation and slows the reactions that the acid drives on the antibody. In the designs the A-Mab concentration is held at its set-point. In the univariate study the concentration is set to each of its planned values before the pH is adjusted.

The acid and base solutions and the other inputs of the step are prepared to their platform specifications. They are controlled for identity and quality and are not characterized in this plan.

5.3 Analytical methods

Each output is measured by a validated method under its own procedure. Table 5.1 lists the method for each output and the sample it is measured in. Size exclusion

chromatography (SEC) is run under SOP-3011, non-reduced capillary electrophoresis with sodium dodecyl sulfate (nrCE-SDS) under SOP-3015, and imaged capillary isoelectric focusing (icIEF) under SOP-3013. The antibody concentration of the load is measured by absorbance at 280 nm (AMV-3019). The log reduction factor (LRF) for XMuLV is the reduction in $\log_{10}$ infectious virus between the spiked pool at the start of the hold and the spiked pool at the end of the hold.

Table 5.1: Analytical methods of the low-pH viral inactivation step, for each output and for the antibody concentration of the load.

Output	Sample	Method	Validation
XMuLV log reduction factor	Spiked pool at the start and at the end of the hold	XMuLV Infectivity Titre (TCID50)	AMV-3017
Aggregate (HMW)	Input pool and neutralized pool	Size-Variants (SEC-HPLC)	AMV-3011
Fragments (LMW)	Neutralized pool	Purity (non-reduced CE-SDS)	AMV-3020
Acidic variants	Input pool and neutralized pool	Charge Variants (icIEF)	AMV-3013
A-Mab concentration	Input pool	Protein Concentration by A280 (UV)	AMV-3019

A-Mab concentration is a parameter of the step and is measured in its load.

From the method decision AMD-001 of 2026-03-20, nrCE-SDS is the method for fragments wherever fragments are measured (PCMP-001 §7.8). Before that date fragments were measured by SEC, and SEC remains the method for aggregate. In the Protein A eluate and the neutralized pool, part of the host cell protein elutes in the low molecular weight window of SEC and is counted as fragment. SEC therefore reads fragments high in the Protein A eluate and the neutralized pool (INV-001 finding 6). Table 5.2 shows which method measured fragments in which design. Aggregate was measured by SEC in both designs.

Table 5.2: Methods used for size variants in the designed studies, low-pH viral inactivation.

Attribute	Method used in study	
	Screening	Response surface
Aggregates, HMW	SEC	SEC
Fragments, LMW	SEC[a)]	SEC[a)], nrCE-SDS[b)]

[a)] In the Protein A and low-pH viral inactivation pools part of the host cell protein elutes in the SEC low molecular weight window. Reported for information only (AMD-001).
[b)] Retained samples of the response-surface design, tested 2026-04-06 to 2026-04-17.

The screening design (S-VI-SCR) and the response-surface design (S-VI-RSM) ran before the method decision and measured fragments by SEC. The screening samples

were not retained, so the screening design has SEC results for fragments only. These results are kept in the record and are not used for any decision. The retained samples of the response-surface design are tested by nrCE-SDS in the re-test study S-VI-RET, and fragments are judged on those results. The qualification study and the univariate study do not measure fragments.

5.4 Sampling plan

Table 5.3 lists the runs of each study, the samples taken and the tests made on them. Each run of a design gives three samples from the spiked run and two from the unspiked run. The sample of the input pool is taken before the pool is acidified, and it gives the level of each attribute that the step starts from. The sample of the neutralized pool is taken after neutralization and filtration, and it gives the level the step delivers to cation exchange.

Table 5.3: Sampling plan of the studies of the low-pH viral inactivation step.

Study	Runs or conditions	Sample	Tests
Qualification of the scale-down model (S-VI-SDM)	Three scale-down runs at set-point and five manufacturing-scale batches	Neutralized pool	Aggregate and acidic variants
Screening design (S-VI-SCR)	11 runs, 3 of them centre points	Input pool, from both runs	Aggregate, acidic variants and A-Mab concentration
		Spiked load, at the start of the hold	XMuLV titre
		Spiked pool, at the end of the hold	XMuLV titre
		Neutralized pool, from the unspiked run	Aggregate, acidic variants and fragments by SEC
Response-surface design (S-VI-RSM)	18 runs, 4 of them centre points	Input pool, from both runs	Aggregate, acidic variants and A-Mab concentration
		Spiked load, at the start of the hold	XMuLV titre
		Spiked pool, at the end of the hold	XMuLV titre
		Neutralized pool, from the unspiked run	Aggregate, acidic variants and fragments by SEC, with the samples retained
Univariate study (S-VI-UNI)	Settings across the range studied, the other three parameters at set-point	Spiked and unspiked runs, as for the designs	XMuLV titre, aggregate and acidic variants
Re-test (S-VI-RET)	Samples retained from the 18 response-surface runs	Neutralized pool	Fragments by nrCE-SDS

Each run of a design is made as a spiked run, which gives the XMuLV titres, and an unspiked run, which gives the quality attributes. The input pool is sampled before acidification and the neutralized pool after neutralization and filtration.
SEC, size exclusion chromatography. nrCE-SDS, non-reduced capillary electrophoresis with sodium dodecyl sulfate.

The centre points are replicate runs at the design centre. They give the pure error of each design, against which lack of fit is tested. The input and the output titre of XMuLV are measured in the same spiked run, so that each run of a design gives its own LRF.

5.5 Statistical methods

The designs and their analysis follow SOP-1002 and PCMP-001 §6. Each factor is coded −1 at the low end of its range studied, +1 at the high end and 0 at the midpoint (Montgomery 2017). An effect is the change in the response over the coded range of the factor, which is twice its coefficient in coded units. A model term is significant when its p-value is below $\alpha = 0.05$.

The screening data will be fitted by least squares with a model of main effects and two-factor interactions. For three factors this model has seven terms. The 11 runs of the design leave four residual degrees of freedom. The three centre points give two of them for pure error, and the other two test lack of fit, which combines curvature and the three-factor interaction. The screening model identifies effects. Its centre points can detect curvature but cannot attribute it to a factor, and no prediction will be made from it.

The response-surface data will be fitted with a full quadratic model, which adds a squared term for each factor to the main effects and the two-factor interactions. The model has ten terms. The 18 runs of the design leave eight residual degrees of freedom, three for pure error from the four centre points and five for lack of fit. The response-surface model is the predictive model. Every prediction, every PAR and the verification of the platform worst cases in PCR-006 will come from it.

Each response-surface model will be reported with R^2, adjusted R^2 and predicted R^2, the model F-test and the lack-of-fit test against pure error. Predicted R^2 is computed from the prediction error sum of squares (PRESS), in which each run is predicted by the model fitted without it. Residuals will be plotted against the predicted values and the run order, and on a normal probability plot, and the Cook’s distance of each run will be computed. The criteria that a model must meet, including the tests on its residuals, are in Section 7.1.

Aggregate, fragments and acidic variants must meet their criteria in every batch, and a response-surface model predicts the mean response. These three attributes are therefore judged through a 95 % prediction interval of a single batch wherever a judgement supports a claim. There are three such judgements. They are the PAR computed while the other factors vary as in routine operation (Section 8), the judgement over the NOR in the qualification of the scale-down model (Section 5.1), which uses the exact prediction interval of a single run, and the verification of the platform worst cases (Section 7.4). The PAR analysis that holds the other parameters at their set-points judges the mean prediction (PCMP-001 §6.4). XMuLV clearance is judged on the mean predicted LRF at the least favourable combination of inactivation pH, hold time and temperature within their NORs, because viral clearance is validated at worst-case conditions. Predictions of aggregate and acidic variants are translated to manufacturing scale by subtracting their offsets (Section 5.1) before they are compared with a criterion. Predictions of fragments and XMuLV clearance are compared as the model gives them.

The results of the univariate study are compared with the set-point runs of the qualification study and with the criteria for aggregate and acidic variants in Table 7.1,

and its log reductions are reported beside the platform claim of the step. A difference from the set-point runs within the precision of the method is attributed to the assay.

6 Study design

6.1 Factors, ranges and study type

Table 6.1 lists the parameters of the step with their planned set-points and NORs and their ranges studied. The range studied is the characterized range of PCMP-001. The code letters identify the factors in the design matrices of Appendices A and B.

Table 6.1: Parameters of the low-pH viral inactivation step, with the set-points and NORs against which the studies were planned.

Code	Parameter	Unit	Planned set-point	Range studied	Planned NOR	Study type
A	Inactivation pH	pH	3.5 [a)]	3.2–4.0	3.4–3.6 [b)]	multivariate
B	Hold time	min	90 [c)]	60–180	60–120 [d)]	multivariate
C	Temperature	°C	21	15–25	19–23 [e)]	multivariate
–	A-Mab concentration	g/L	20	5–35	5–35 [f)]	univariate

[a)] PCR-006 Version 2.0 moved the set-point to pH 3.4.
[b)] PCR-006 Version 2.0 moved the NOR to pH 3.3–3.5.
[c)] PCR-006 Version 2.0 moved the set-point to 110 min.
[d)] PCR-006 Version 2.0 narrowed the NOR to 100–120 min.
[e)] PCR-006 Version 2.0 narrowed the NOR to 20–23 °C.
[f)] The NOR equals the range studied.
Code, the letter of the factor in the design matrices of Appendices A and B. A-Mab concentration is not a factor of the designs.

For inactivation pH and temperature, the range studied extends beyond the NOR at both ends. It shows how the step behaves near and beyond the edges of routine operation, and it bounds the knowledge space within which a PAR can be claimed. For hold time, the range studied starts at the lower end of the NOR. That end is the shortest hold that the NOR of this plan allows, and the least favourable condition for XMuLV clearance. The range extends beyond the upper end of the NOR to show how aggregate and fragments grow in longer holds. For the A-Mab concentration the NOR equals the range studied.

The centre of each design is the midpoint of the ranges studied, at pH 3.6, 120 min and 20 °C. It is not the set-point of any of the three factors, because the ranges are not symmetrical about the set-points. The centre points therefore estimate pure error at the design centre. Each model predicts the outputs at the set-point from the fitted surface.

6.2 Screening design

The screening design (S-VI-SCR) is a two-level full factorial design in the inactivation pH, hold time and temperature, with three centre points, in 11 runs. Every combination of the low and high ends of the three ranges is run, so the design estimates all main effects and all two-factor interactions without aliasing. Its purpose is to identify which factors and interactions act on each output before the response-surface design is run. All three factors are carried into the response-surface design, because each of them is expected to act on XMuLV clearance (Section 4.3). The runs are executed in random order. The planned design is in Appendix A.

6.3 Response-surface design

The response-surface design (S-VI-RSM) is a face-centred central composite design in the same three factors (Montgomery 2017). It has eight factorial runs, six axial runs and four centre points, 18 runs in all. The axial runs lie at the centres of the faces of the design cube, at the ends of each range studied, so every run stays inside the range studied. Together with the centre points, the axial runs allow a squared term to be estimated for each factor, which the full quadratic model needs. The runs are executed in random order. The planned design is in Appendix B.

Every output in Table 5.1 is measured on every run. Fragments are judged on the nrCE-SDS results of the retained samples (Section 5.3).

6.4 Univariate assessment

The A-Mab concentration is assessed one factor at a time in the univariate study (S-VI-UNI). The low-pH hold is run at settings of the concentration across the range studied, 5–35 g/L, with the inactivation pH, hold time and temperature held at their set-points. Aggregate, acidic variants and the log reduction of XMuLV are measured on every run, and the set-point runs of the qualification study (S-VI-SDM) serve as the reference at set-point. The study does not measure fragments, so it supports no range of the concentration for that attribute, and Section 12 states the assumption the plan then rests on. A study of one factor at a time cannot show whether the effect of the concentration depends on the other parameters.

7 Acceptance and decision criteria

The criteria below apply PCMP-001 §7 to this step. They are derived from the drug substance criteria, the platform history of X-Mab, Y-Mab and Z-Mab, and the bounds of downstream steps verified on the data of this campaign (PCMP-001 §7.3 and §7.5). No result of the studies in this plan enters them. PCR-006 will report each result against its criterion and state whether the criterion was met.

7.1 Scale-down model and model adequacy

1. The scale-down model is qualified for aggregate and for acidic variants when the conclusion against the criterion over the NOR is the same with the offset at either end of its 90 % confidence interval (Section 5.1). The decision is made at the analysis, with the response-surface model.
2. A response-surface model is adequate for prediction when all four of the following conditions hold (PCMP-001 §7.2).
   - The model F-test is significant at $\alpha = 0.05$.
   - Lack of fit against pure error is not significant at $\alpha = 0.05$.
   - Predicted R^2 falls short of R^2 by no more than 0.2, the limit on overfitting of PCMP-001 §7.2.
   - The residuals show no trend, and no run dominates the fit. A regression of the absolute residuals on the predicted values, and a regression of the residuals on run order, each has no slope significant at $\alpha = 0.05$. No run has a Cook's distance above 4/n, where n is the number of runs in the fit (0.22 for the 18 runs of the response-surface design).
3. A model that fails the lack-of-fit test or the limit on overfitting is either refitted with a justified change of terms or transformation, or it is not used to set a range (PCMP-001 §7.2). This plan sets no rule for which terms a refit removes or which transformation it applies, so PCR-006 states the change it made to any model it refitted, with the reason for it.
4. Where no adequate model results for an output, no range is set from the model of that output. PCR-006 then reports the output descriptively against its criterion, over the runs of the response-surface design (PCMP-001 §7.2).

7.2 Intermediate acceptance criteria

A drug substance criterion does not say what the output of an upstream step must meet. Each output of this step is therefore judged against an intermediate accep-

tance criterion, as PCMP-001 §7.3 derives it. The drug substance criterion is carried backward through the train using the worst case that each downstream step showed on the platform. That worst case is the lowest clearance factor for a step that removes the attribute and the largest increase for a step that adds to it. No margin is applied. An attribute that no downstream step changes keeps its drug substance criterion. Table 7.1 gives the criteria for the outputs of this step.

Table 7.1: Intermediate acceptance criteria for the low-pH viral inactivation output and their derivation from the drug-substance criteria.

Attribute	DS criterion	Downstream worst case (platform or verified)		IAC	Basis
		Clearance factor	Increase		
Aggregates, HMW (%)	≤ 5.00	1.40	0.00	≤ 7.02	Carried back, tightened[a)]
Fragments, LMW (%)	≤ 2.00	1.30	0.00	≤ 2.60	Carried back
Acidic variants (%)	18.0–40.0	1.00	0.00	18.0–40.0	DS criterion[b)]
XMuLV clearance (log10)	≥ 16.7[c)]	not applicable		≥ 6.60	Platform claim[d)]

[a)] The response-surface model of a downstream step predicted, within its NOR, a worse value than the platform worst case. The one-sided 95 % prediction bound of a single batch there replaced it.
[b)] No downstream step changes the attribute, so the drug-substance criterion applies.
[c)] Cumulative requirement across the steps that clear the virus.
[d)] Log reduction claimed for this step in the modular viral clearance claim.
DS, drug substance. IAC, intermediate acceptance criterion. Increase, in the unit of the criterion.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

In Table 7.1 the downstream worst case is the cumulative clearance factor and increase of the steps after this one.

Aggregate in the neutralized pool must not exceed 7.02 % HMW. Cation exchange is the only step after this one that is credited with removing aggregate, so the criterion is the drug substance criterion multiplied by the cation exchange clearance factor. Within its NOR, cation exchange cleared less aggregate than its platform worst case (PCR-007). The one-sided 95 % prediction bound of the clearance factor of a single cation exchange batch, at the least favourable combination within its NOR, therefore replaced the platform value (PCMP-001 §7.5). The a priori criterion of 10.0 % HMW tightened to the value in Table 7.1.

Fragments in the neutralized pool must not exceed 2.60 %, measured by nrCE-SDS. This criterion is the drug substance criterion carried back through the platform worst case of cation exchange, the only step after this one that is credited with removing fragments. Acidic variants must lie within 18.0–40.0 %, the drug substance criterion, because no downstream step is credited with a change in them. Both limits of the

acidic variant criterion apply. XMuLV clearance must reach the platform claim of the step (Section 7.3).

7.3 XMuLV clearance claim

Each step that clears a virus is held to its platform claim, which is the log reduction that the step has delivered for that virus on the platform (PCMP-001 §7.4). For XMuLV the claims are 6.6 $\log_{10}$ for this step, 5.5 $\log_{10}$ for anion exchange and 4.62 $\log_{10}$ for virus filtration. They sum to 16.72 $\log_{10}$, which exceeds the cumulative requirement of 16.7 $\log_{10}$ by 0.02 $\log_{10}$. A shortfall at this step larger than that surplus therefore cannot be made up by the claims of the other two steps. Anion exchange and virus filtration are characterized under PCP-008 and PCP-009, and PCMR-001 consolidates the cumulative claim.

The XMuLV clearance of this step is judged at the worst case within the NOR. The lowest mean predicted LRF among all combinations of NOR low, set-point and NOR high of the inactivation pH, hold time and temperature must reach 6.6 $\log_{10}$.

7.4 Verification of the platform worst cases

The criteria upstream of this step assume that the low-pH step adds no more aggregate, fragments and acidic variants than it added on the platform. Table 7.2 lists these platform worst cases and the criteria upstream that rely on them. PCR-006 will check each of them with the response-surface model of this step (PCMP-001 §7.5).

Table 7.2: Platform worst cases of the low-pH viral inactivation step that criteria upstream of it rely on.

Attribute	Increase (percentage points)	Upstream criterion
Aggregates, HMW (%)[a)]	0.70	Aggregate, production bioreactor output
Fragments, LMW (%)[b)]	0.15	Fragments, Protein A output
Acidic variants (%)	1.00	Acidic variants, production bioreactor output

[a)] The platform value also draws on Table 4.16 of the A-Mab case study.
[b)] Measured by nrCE-SDS.
Increase, the largest increase in the attribute across the step for X-Mab, Y-Mab and Z-Mab, over a hold of 180 min.
Upstream criterion, the criterion of a step before this one that relies on the platform worst case in the same row.

The verification of the platform worst cases takes the largest increase that the response-surface model predicts within the NOR of Table 6.1, at the least favourable combination of NOR low, set-point and NOR high. It also takes the one-sided 95 %

prediction bound of a single batch at that combination. The bound is a prediction bound of a single batch because each output is judged per batch, and the platform worst case that it may replace was a single run. An assumption is confirmed when the predicted largest increase is no larger than the platform worst case. The results observed at the ends of the range studied are reported beside the predicted largest increase, for information.

The platform worst cases in Table 7.2 are increases over a hold of 180 min, the upper end of the range studied. The verification takes the A-Mab increase within the NOR, where the hold is at most 120 min. A bound smaller than a platform value may therefore reflect the shorter hold of the A-Mab NOR as well as the behaviour of the product, and PCR-006 states which of the two it reflects, for each attribute, from the runs made at a hold of 180 min. The criterion that results is still right, because routine manufacture never holds the pool longer than the upper end of the NOR. The verification may change the criteria upstream of this step by the following rule.

1. Where the largest increase predicted within the NOR exceeds the platform worst case, the 95 % prediction bound of a single batch replaces the platform value. Every criterion upstream of this step is recalculated with the bound and becomes stricter.
2. Where the 95 % prediction bound of a single batch is smaller than the platform worst case, and the response-surface model meets the adequacy criteria of Section 7.1, the bound replaces the platform value. Every criterion upstream of this step is recalculated with the bound and becomes wider.
3. Otherwise the platform value stands.

Where the rule widens a criterion, PCR-006 should give the argument for it, with the bound, the runs behind it and the reason why the performance of this step for A-Mab is known well enough to replace the platform value. Every change of a criterion, in either direction, should also be argued by its risk. The argument names who bears the risk, whether and where an excursion would be detected, what the downstream steps must then clear and the probability of a drug substance out of specification.

7.5 Parameter classification

PCR-006 will classify each parameter by the rule of PCMP-001 §7.7, which reads as follows.

> Parameters are classified under the parameter classification procedure SOP-4001 by the decision logic of the A-Mab case study (CMC Biotech Working Group 2009).
>
> 1. A critical process parameter (CPP) has an effect on a CQA that is significant and practically relevant within the characterized range, and a high risk of falling outside its acceptable range in operation.

2. A well-controlled critical process parameter (WC-CPP) has the same kind of effect on a CQA but a low risk of falling outside its acceptable range, because it is readily controlled and its PAR is wide against its NOR.
3. A key process parameter (KPP) affects process performance, such as yield or concentration, and has no effect on a CQA within its range.
4. A general process parameter (GPP) has no relevant effect on a CQA or on process performance within its range.

A significant effect that is too small to move the attribute appreciably within the PAR does not by itself make a parameter critical. The PCR-00N should state the size of the effect beside its significance and give the reason for each class.

An effect is significant when a term of the response-surface model that contains the parameter has a p-value below $\alpha = 0.05$ (PCMP-001 §6.2). Neither PCMP-001 nor SOP-4001 sets a numerical threshold for an effect that is practically relevant, or for a PAR that is wide against its NOR, so neither decides a class on its own. PCR-006 states, for each parameter and each attribute it affects, the size of the effect across the PAR and the margin between the PAR and the NOR, and gives the reason for each class. Criticality is treated as a continuum, as described in ICH Q8(R2) (International Council for Harmonisation 2009).

8 Proven acceptable ranges (planned analysis)

A PAR will be computed for each attribute that has a criterion at this step and for each factor of the response-surface model. Two analyses will be run (PCMP-001 §6.4). The first holds the other parameters at their set-points and scans the parameter of interest across its range studied. Its PAR is the contiguous range around the set-point over which the mean prediction meets the criterion.

The second analysis computes the PAR while the other two factors of the response-surface model vary about their set-points as in routine operation. PCMP-001 §6.4 calls it the NOR-propagated analysis. It scans the parameter across 81 grid points over its range studied. How the other factors vary, and what is compared with the criterion, depend on the attribute.

- For aggregate, fragments and acidic variants it compares the 95 % prediction interval of a single batch, because a single batch is what must meet the criterion. At each grid point, 2,000 operating points are drawn for the other two factors. Each factor is drawn from a normal distribution centred on its set-point, with a standard deviation of one sixth of the width of its NOR, so that the NOR spans three standard deviations on either side of the set-point. A draw that falls outside the range studied is set to the nearer end of that range. The interval holds the central 95 % of the simulated batches, and it includes the model residual and the uncertainty of the fitted coefficients. For an upper limit the upper end of the interval must not exceed the limit, for a lower limit the lower end must not fall below it, and for a two-sided criterion both apply.
- For XMuLV clearance it compares the worst case within the NOR. At each grid point the other two factors are set to every combination of their NOR low, set-point and NOR high, and the lowest mean predicted LRF must reach the claim.

The PAR of the second analysis is the contiguous range around the set-point over which the comparison passes. It is in general narrower than the PAR of the first analysis, because the range must hold while the other factors vary. The PAR of the second analysis is the one that supports a claim. The decision rules are the following.

1. A range of a parameter is proven acceptable for an attribute when the second analysis meets the criterion of that attribute across it.
2. A parameter that affects several attributes has a PAR for each attribute, and the range carried into the control strategy is their intersection.
3. A PAR that reaches the end of the range studied is reported as reaching it, and no claim is made beyond it.

4. Where an output does not meet its criterion even at the set-point, PCR-006 reports that no PAR exists for that attribute and states the consequence for the NOR.
5. The range of the A-Mab concentration is supported by its own univariate data. It is supported when every result across the settings of the concentration meets the criterion for aggregate and for acidic variants in Table 7.1, at manufacturing scale. PCR-006 reports the log reduction measured at each setting beside the claim of the step, which is itself judged on the response-surface model at the worst case within the NOR (Section 7.3). The univariate study supports no range of the concentration for fragments (Section 12).

PCR-006 will tabulate both PARs for each attribute and each parameter. For each attribute it will also show a figure against the parameter with the largest main effect in the response-surface model. The figure plots the prediction across the range studied, with the region that meets the criterion shaded green.

9 Data management and integrity

The data of these studies are recorded and kept to the ALCOA+ principles. Raw data from the analytical instruments and the scale-down systems are held in validated electronic systems with audit trails. Records made by hand are made on controlled forms. Every study record carries the execution window of the study and the calibration status of the equipment it used (INV-001 finding 5).

Results are transferred to the statistical analysis from the approved record, and a second analyst verifies the transfer. A result is excluded only for a documented reason, and PCR-006 lists every exclusion. Where a run is lost or invalidated, PCR-006 should report it and state whether the design was repaired or re-executed (PCMP-001 §6.1).

10 Roles and responsibilities

The functions that approve this plan share the work of the studies as Table 10.1 shows.

Table 10.1: Responsibilities of the functions that approve this plan.

Function	Responsibility
Process Development / MSAT	Author of this plan and of PCR-006. Runs the scale-down model, the designs and the univariate study, and analyses the data.
Analytical Development	Runs the SEC, nrCE-SDS and icIEF methods and the XMuLV infectivity assay under their method validations. Owns the method decision AMD-001.
Manufacturing Science	Supplies the manufacturing-scale data for the qualification of the scale-down model, and transfers the results to the receiving site.
Statistics	Reviews the designs, fits the models, and runs the PAR analysis and the verification of the platform worst cases.
Quality Assurance	Approves this plan and PCR-006, and checks that every criterion applied in PCR-006 is the one stated in this plan.

11 Deliverables and schedule

The deliverable of this plan is the paired Process Characterization Report PCR-006. PCR-006 reports every study in Table 11.1 against the criteria of Section 7, and it rolls up into PCMR-001 with the reports of the other steps. The qualification runs of the scale-down model are made before the first characterization run (PCMP-001 §5). The qualification decision needs the response-surface model, so it is made at the analysis, after the last study. The screening design is run before the response-surface design, so that its effects are known when the response-surface design is analysed. The re-test by nrCE-SDS follows the method decision AMD-001.

Table 11.1 also records the execution window of each study, which the Version 1.0 record did not (INV-001 finding 5). The first study of the step started on 2026-01-07, after this plan was approved, and the response-surface design and the univariate study of A-Mab concentration ran in parallel.

Table 11.1: Sequence of the studies under this plan and the execution window of each.

Phase	Study	Content	Start	End
1	S-VI-SDM	Set-point runs of the scale-down model, compared with the manufacturing-scale batches, and the offsets	2026-01-07	2026-01-16
2	S-VI-SCR	Screening design	2026-01-19	2026-01-30
3	S-VI-RSM	Response-surface design	2026-02-02	2026-02-13
	S-VI-UNI	Univariate study of A-Mab concentration	2026-02-02	2026-02-06
4	S-VI-RET	Re-test by nrCE-SDS of the retained samples of the response-surface design	2026-04-06	2026-04-17

Start, the date of the first run or test of a study. End, the date of the last. Studies in one phase ran in parallel.
The qualification decision, the analysis of all studies and the report PCR-006 follow the last study.

12 Risks and assumptions

The studies carry the following risks.

1. The scale-down model may differ in mean from manufacturing scale for aggregate and acidic variants. The offsets and the qualification rule of Section 5.1 address this, and every conclusion for these two outputs is stated at manufacturing scale.
2. Fragments and XMuLV clearance are not compared at both scales, so their offsets are unknown (Section 5.1). Their judgements rest on the spiking model and on the assumptions below.
3. An infectivity titre carries the variability of a cell-based assay. The XMuLV claim is therefore judged on the mean prediction of the response-surface model and not on a single titre. The centre points estimate the variability of the LRF at the design centre.
4. The design centre is not the set-point (Section 6.1). The models predict the outputs at the set-point and do not measure them there. Replicate runs at set-point are made only in the qualification study.
5. The screening design has no fragment results that can be used for a decision (Section 5.3). Effects on fragments are therefore estimated from the response-surface design only.
6. The designs and the univariate study need Protein A eluate produced at the Protein A set-points. A shortage of eluate would delay the studies, but it would not change the designs.

The plan rests on the following assumptions.

1. Each downstream step performs at least as well as the platform worst case on which the criteria of this step rely. Each such assumption is checked in the report of that step. The verification of the cation exchange aggregate clearance has already changed the aggregate criterion of this step. Cation exchange cleared less aggregate within its NOR than its platform worst case, and the one-sided 95 % prediction bound of a single batch for that clearance is part of the aggregate criterion (Section 7.2).
2. The step does not change the glycan attributes or host cell protein, and no credit is taken for either.
3. The scale-down model forms fragments at the rate of manufacture, because cleavage in the hinge region depends on pH, time and temperature and not on scale (Section 5.1).
4. The A-Mab concentration within its range studied does not change the fragments formed in the hold. The designs hold the concentration at its set-point, and the univariate study does not measure fragments (Section 6.4). Cleavage in the

hinge region is a reaction of each molecule with the solvent, so the fraction of fragments formed is not expected to depend on the antibody concentration.

5. Each NOR spans the routine variation of its parameter, as the second PAR analysis assumes (Section 8).

13 Approvals

This plan is approved by the functions listed in the approval record at the front of this document. A change to the plan after approval is made in a new version, approved by the same functions.

Appendix A. Planned screening design

Table A.1 gives the planned screening design (S-VI-SCR), with the coded level and the setting of each factor.

Table A.1: Planned screening design of the low-pH viral inactivation step.

Run type	Run	Coded level			Setting		
		A	B	C	pH	Hold time (min)	Temperature (°C)
Factorial	1	−1	−1	−1	3.2	60	15
	2	−1	−1	+1	3.2	60	25
	3	−1	+1	−1	3.2	180	15
	4	−1	+1	+1	3.2	180	25
	5	+1	−1	−1	4.0	60	15
	6	+1	−1	+1	4.0	60	25
	7	+1	+1	−1	4.0	180	15
	8	+1	+1	+1	4.0	180	25
Centre point	9	0	0	0	3.6	120	20
	10	0	0	0	3.6	120	20
	11	0	0	0	3.6	120	20

A, inactivation pH. B, hold time. C, temperature.
The coded levels −1, 0 and +1 are the low end, the midpoint and the high end of the range studied.
Runs are listed in the order in which the design was generated, which is not standard order. They are executed in random order.

Appendix B. Planned response-surface design

Table B.1 gives the planned response-surface design (S-VI-RSM), with the coded level and the setting of each factor.

Table B.1: Planned response-surface design of the low-pH viral inactivation step.

Run type	Run	Coded level			Setting		
		A	B	C	pH	Hold time (min)	Temperature (°C)
Factorial	1	−1	−1	−1	3.2	60	15
	2	−1	−1	+1	3.2	60	25
	3	−1	+1	−1	3.2	180	15
	4	−1	+1	+1	3.2	180	25
	5	+1	−1	−1	4.0	60	15
	6	+1	−1	+1	4.0	60	25
	7	+1	+1	−1	4.0	180	15
	8	+1	+1	+1	4.0	180	25
Axial	9	−1	0	0	3.2	120	20
	10	+1	0	0	4.0	120	20
	11	0	−1	0	3.6	60	20
	12	0	+1	0	3.6	180	20
	13	0	0	−1	3.6	120	15
	14	0	0	+1	3.6	120	25
Centre point	15	0	0	0	3.6	120	20
	16	0	0	0	3.6	120	20
	17	0	0	0	3.6	120	20
	18	0	0	0	3.6	120	20

A, inactivation pH. B, hold time. C, temperature.
The coded levels −1, 0 and +1 are the low end, the midpoint and the high end of the range studied.
Runs are listed in the order in which the design was generated, which is not standard order. They are executed in random order.

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